

Economic Weather Industry Rotation

with Active Portfolio Optimization

MGMTMFE 431 Quantitative Asset Management Final Project

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Research Question

Can macro “weather” improve industry rotation?

We test whether a monthly, long-only portfolio across Fama-French industry portfolios can improve risk-adjusted performance by changing its industry tilts across macro regimes.

Core intuition

Industries are economic habitats: defensives, extractors, cyclicals, and technology networks should respond differently to inflation, rates, labor markets, and industrial growth.

Backtest design

- **Universe:** 47 aligned FF industry portfolios
- **Benchmarks:** equal-weight industries and market
- **Window:** 120-month rolling estimation
- **Rebalance:** monthly
- **OOS:** Jan 1990–Oct 2025

Data & Forecast Inputs

Return and factor data

- FF 49 industry monthly returns
- FF five factors and momentum
- AQR QMJ and BAB factors
- Risk-free rate from FF data

Macro weather variables

- Term spread and short rates
- Inflation
- Industrial production growth
- Unemployment change

Macro variables are lagged at portfolio formation to avoid look-ahead bias.

Forecast components

- Shrunk industry alpha from rolling factor regressions
- Factor-premia forecast using historical average and lagged macro predictors
- 12-month industry momentum, skipping the latest month
- Weather regime indicator

All predictors are lagged before portfolio formation to avoid look-ahead bias.

Economic Weather Overlay

State	Condition	Tilt
Sunshine	Strong IP, stable labor	Cyclicals, tech
Goldilocks	No macro pressure	Networks, household beta
Heatwave	High inflation, rising rates	Extractors, utilities
Storm	Inversion or rising unemployment	Defensive shelter

How weather enters the model

- Adds an economic prior to expected returns
- Raises caps for favored industries
- Tightens caps for disfavored industries
- Scales risk aversion by regime

Risk appetite

Supportive regimes allow more active risk; storm regimes make the optimizer more conservative while remaining fully invested.

Portfolio Construction

Expected return forecast

$$\hat{\mu}_{i,t} = \tilde{\alpha}_{i,t} + \hat{\beta}'_{i,t} \tilde{f}_t + 0.012 z(\text{MOM}_{i,t}) + \omega_{i,t}^{\text{weather}}$$

Risk model

$$\hat{\Sigma}_t = \hat{B}_t \hat{\Sigma}_{f,t} \hat{B}_t' + \hat{D}_t$$

Active optimization

$$\max_w w' \hat{\mu}_t - \frac{\gamma_t}{2} (w - b)' \hat{\Sigma}_t (w - b)$$

$$\sum_i w_i = 1, \quad 0 \leq w_i \leq c_{i,t}$$

Weather-dependent controls

Favored industries can receive higher caps; disfavored groups get tighter caps. Risk aversion is scaled by weather state.

Main Result

Portfolio	Return	Vol.	Sharpe	MDD
Weather strategy	11.8%	15.2%	0.77	−43.5%
Equal-weight	9.1%	16.0%	0.57	−54.0%
Market	8.9%	15.2%	0.59	−54.1%
10-ind. robust	10.6%	14.4%	0.74	−39.8%

Takeaway

The strategy improves Sharpe and drawdown while staying long-only and fully invested in equity industries.

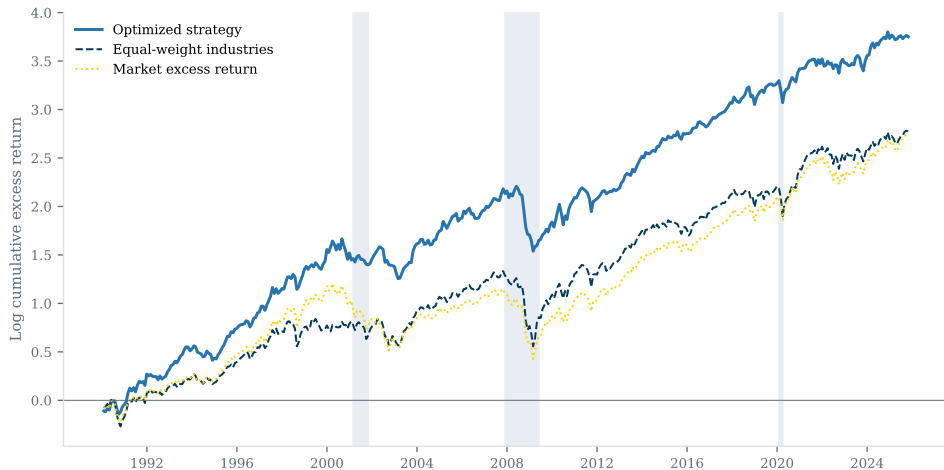
Active return vs. equal-weight:

+2.69% ann.

Tracking error: **9.08%**

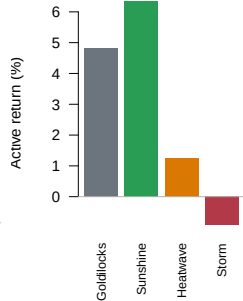
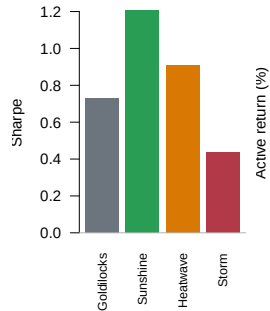
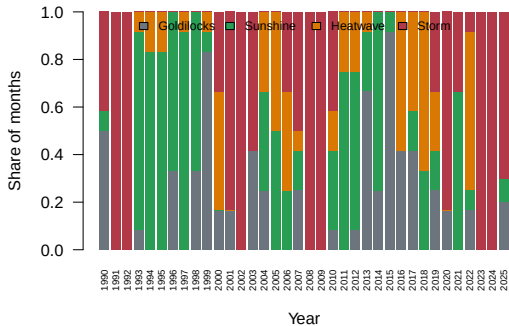
Information ratio: **0.30**

Cumulative Performance



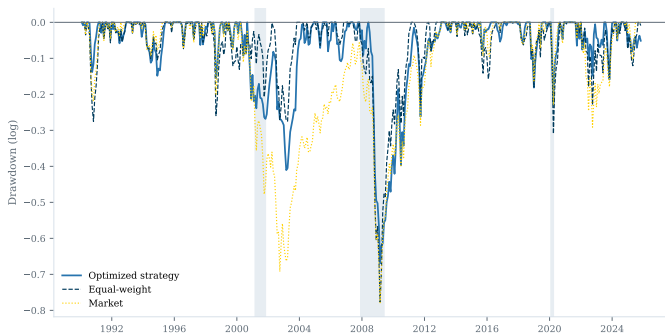
The strategy compounds faster than both benchmarks while still sharing broad equity-market risk.

Weather Regimes



Weather states are economically interpretable: supportive regimes perform best, while storm months remain difficult for equity-only portfolios.

Risk & Factor Interpretation



Drawdowns remain equity-like, but are less severe than both benchmarks.

Factor-adjusted result

Term	Loading	t-stat
Alpha	0.05%	0.60
Mkt-RF	0.98	38.3
Momentum	0.29	13.3
RMW	0.28	4.8
BAB	0.07	2.6

Interpretation

The strategy is best framed as disciplined allocation across economic habitats, not as standalone alpha.

Implementation Checks

Turnover and costs

Average one-way turnover is **20.1% per month**.

Cost	Net return	Sharpe
0 bps	11.75%	0.77
10 bps	11.42%	0.75
25 bps	10.92%	0.72

Subperiod Sharpe

Period	Strategy	EW	Mkt
1990s	1.12	0.63	0.93
2000s	0.34	0.27	−0.10
2010s	1.08	0.86	1.01
2020–25	0.58	0.59	0.74

The strategy improves long-run performance, but does not remove regime dependence.

Conclusion

What we find

- Weather-aware rotation improves Sharpe: **0.77** vs. 0.57 equal-weight
- Drawdown is smaller, but still equity-like
- Transaction costs reduce, but do not erase, the edge

Important caveat

- Factor alpha is not statistically significant
- The contribution is disciplined allocation across macro habitats, not a pure anomaly